Slip Face Detection and Route Planning for Small Rovers on Sand Dunes, Tested Against Airborne Lidar

Huda Mueen
Independent Researcher, United Arab Emirates
hudamueen2407@gmail.com

***Abstract*—For most deserts the only free elevation model a small wheeled rover can plan on has a posting of about 30 m, while the main hazard on active dunes, the slip face, is often only metres long. We built a benchmark on six North American dune fields, with one metre airborne lidar as truth, to measure what coarse maps hide and what a planner can do about it. Ten families of hazard layers, from slope and interpolation to boosted trees, a convolutional network, learned super resolution and imagery fusion, were scored against a lidar face map on 10 and 30 m lidar block means (four of them from 2 to 60 m) and on three real elevation products, and were used by a slip aware planner at operating points chosen on held out sites. At 10 m the best layer raises mission success from 17 % to 64 %; at 30 m no layer we planned with adds more than 5 points to 8 %, because most faces are shorter than one cell. A face map from fine data would reach 81 % on the 30 m grid, and onboard sensing that sees every face within 20 m keeps the rover off faces whatever the prior. In post hoc checks, the face map fell from 84 % to 61 % when displaced by 10 m, and with line of sight the rover succeeded on 30 % of the pairs with the slip model prior and 74 % with a face map prior. Code, results and the prediction log are public.**

***Index Terms*—field robotics, traversability, digital elevation models, sand dunes, slip faces, path planning, benchmark**

I. Introduction

Small wheeled ground robots are proposed for survey, inspection and search in sand deserts, including the dune seas of the Arabian Peninsula. Before such a robot drives, its route is planned on a prior map, and for most deserts the only free elevation model is a global product with a posting of about 30 m [1]. The dominant hazard on active dunes is the slip face, the lee side of a dune held near the angle of repose of dry sand, about 30 to 34 degrees [2], where wheel slip rises until a small vehicle can no longer climb [3], [4]. Many slip faces are shorter than 30 m (Section IV-B), so a 30 m cell can hold one while its mean slope is gentle.

That a coarse grid understates slope is long known [5]. What is not known is how much this matters for a planner, which hazard layer a practitioner should compute from a coarse map, how fine a map must be, and whether imagery, a different product or onboard sensing change the answer. This paper answers these questions with a benchmark on six North American dune fields where one metre airborne lidar provides the truth.

The contributions are: (1) a benchmark that scores hazard layers against a lidar face map whose meaning is the same at every resolution, on surfaces from 2 to 60 m, on three real elevation products and with Sentinel-2 and NAIP imagery; (2) a comparison of ten families of hazard layers, every learned layer trained leave one site out; (3) a planning evaluation at operating points chosen on the other sites, with degraded face maps and a closed loop with onboard sensing; and (4) a scaling result, detection skill against face length over grid spacing.

II. Related Work

Overhead priors. Vandapel et al. [6] planned ground vehicle routes on airborne lidar collected before the mission, and overhead imagery or lidar at metre scale has served as a cost prior since [7]. Drones can map ahead of the robot [8] or learn terrain properties from the air [9]. Planetary missions plan from orbital data: traversability analysis selected the Mars 2020 landing site [10], and slip predicted from orbital data drives route plans [11]. These priors are at metre scale; we ask what a 10 to 60 m prior, the only free one for most deserts, hides.

Slip on sand. Geometric costmaps score slope and roughness on a grid [12], [13], [14], and recent reviews cover traversability risk [15]. On loose sand slip rises steeply with slope [3], [16]; Curiosity met high slip crossing megaripples [4]; robophysical studies relate slope and granular flow [17]. Planners propagate slip uncertainty into routes [18], [19], [20], but they assume that the map resolves the slope, which fails on dunes at 30 m.

Dunes in elevation models. Coarser grids understate slopes [5]. Dune morphometry from DEMs and imagery is well developed [21], and deep learning maps crestlines from DEM derivatives [22]. Global 30 m products have been ranked against lidar and laser altimetry [23]; TanDEM-X, the source of GLO-30, performs worse over sand [24], and the error of open DEMs in a dune sea grows on slopes above 15 degrees [25]. Learned super resolution can sharpen coarse DEMs [26], [27]. We know of no benchmark that scores coarse map hazard layers and planners for slip faces against lidar truth across resolutions.

III. Benchmark

A. Sites, truth and label

Six windows of USGS 3DEP one metre bare earth lidar [28] cover active dune fields (Table I); we chose them from 20 m slope images and had used them in earlier tests of the detector.

TABLE I
LIDAR WINDOWS (USGS 3DEP ONE METRE DEM; FLOWN: ACQUISITION PERIOD OF THE LIDAR PROJECT), THE NUMBER OF FACE PATCHES AND THE SHARE OF CELLS HOLDING A FACE PATCH POST AT 10 AND 30 M.

Site	Lat., lon.	km^2	Relief	Flown	Patches	10 m	30 m
Dumont Dunes, CA	35.65, -116.27	14.4	322 m	2024 to 25	4693	36 %	49 %
White Sands, NM	32.82, -106.29	6.3	20 m	2020	789	15 %	40 %
Kelso Dunes, CA	34.91, -115.72	9.7	240 m	2024 to 25	2184	15 %	38 %
Great Sand Dunes, CO	37.75, -105.54	14.8	246 m	2019 to 20	2065	16 %	32 %
Algodones Dunes, CA	32.92, -115.10	20.0	122 m	2021 to 23	3131	10 %	27 %
Little Sahara, UT	39.70, -112.39	6.4	99 m	2020 to 21	1776	11 %	35 %

A post is *steep* when its Horn slope [29] at 1 m is 28° or more, a little below the angle of repose of dry sand [2]; voids never count as steep. A *face patch* is an 8 connected group of at least 10 steep posts (10 m^2). Face patches are steep ground, not verified slip faces. For a coarse cell c with footprint F_c the label is the face map

$$y_c = \begin{cases} 1 & \text{if some } x \in F_c \text{ lies in a face patch,} \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

which a planner needs and whose meaning does not change with the cell size. The cell label of the first version of this benchmark, at least 10 steep posts in F_c, means 10 % of a 10 m cell but 1.1 % of a 30 m cell and is reported only for comparison.

B. Inputs

Idealised: block means of the lidar at 2, 5, 10, 15, 20, 30 and 60 m (I10 and I30 in the tables) (no product noise, time gap or misregistration). *Real:* the USGS 1/3 arc second DEM (R10, about 8×10 m) [30], NASADEM (N30, February 2000) [31] and Copernicus GLO-30 (G30, 2011 to 2015) [1], each scored on its own geographic grid, with labels from the lidar posts inside each native cell; no horizontal shift is applied. For planning, GLO-30 is interpolated to the 30 m lidar grid. Over these sites R10 is probably resampled by USGS from the same lidar collections. *Imagery:* one Sentinel-2 L2A scene per site [32] (10 m; lidar year, cloud below 5 %, lowest sun, chosen by rule) and NAIP [33] (0.3 to 0.6 m, resampled to 1 m).

C. Hazard layers

Table II lists the layers, among them a logistic subgrid detector trained only on synthetic dune profiles, which ships with the planner and is one baseline among others. Every learned layer is trained leave one site out [34].

D. Planner

The planner searches the 8 connected grid of the input. For an edge with along track grade $\tan\alpha$ (uphill positive) and cross slope $\tan\beta$,

$$d = \sqrt{(\tan\alpha + f_r)^2 + \tan^2\beta}, \qquad i = -i_0 \ln\left(1 - d/\tan\phi\right) \tag{2}$$

give the traction demand and the slip ($i = \infty$ when $d \geq \tan\phi$), with friction angle ϕ, motion resistance coefficient f_r and slip scale i_0; the slip law is the exponential shear relation of terramechanics [35] written for the ratio of demand to capacity. An edge is passable when $i \leq i_{\max}$ and costs $\ell/(1-i)$ for length ℓ. The four parameters are drawn 128 times; a cell is passable for a draw when at least four of its eight headings are, and q_c is the share of draws for which it is. A cell is BLOCKED when $q_c \leq 0.05$, TRAVERSABLE when $q_c \geq 0.95$ and its face probability $p_c < \tau$, and UNKNOWN otherwise. The main setting, the *posterior vehicle*, draws the parameters from a posterior fitted by Markov chain Monte Carlo to published slip curves of other wheeled vehicles (5 to 95 % ranges: ϕ 30.6 to 43.7°, f_r 0.03 to 0.32, i_0 0.33 to 1.21; $i_{\max}$ 0.4 to 0.7; nominal climbing limit 15.9°); a narrower assumed range (limit 25.5°) is the sensitivity. Three policies share the graph: *slip only*, *gate* (TRAVERSABLE cells only) and *weight* (no BLOCKED cell, cost times $1 + w\,p_c$ of the cell entered); gate and weight refuse descents of 28° or more. Every layer is run at $\tau \in \{0.02, \ldots, 0.9\}$ and $w \in \{1, \ldots, 100\}$ and reported at the setting that maximises mission success on the other five sites.

E. Scoring and closed loop

One hundred pairs per site, at least 1 km apart, are drawn twice: E1 from cells whose lidar posts are all gentle (truth used, as in our earlier runs) and E2 from cells whose coarse slope is below 15° (no truth). A route, the polyline through its cell centres sampled every 0.5 m on the lidar, is *unsafe* when a sample falls on a face patch, and a *mission success* when it exists and is not unsafe. In the closed loop the rover follows the plan on a 5 m grid that inherits the coarse classes and face probabilities ($w = 5$), with slip costs from the coarse surface interpolated to 5 m; at every step it senses the face patches within a radius R from the lidar, blocks those cells, drops the face penalty of sensed free cells and replans when its route is blocked, within a window 300 m around the pair and up to 300 replans.

F. Statistics and prediction log

AUC intervals come from a spatial block bootstrap [36] with 300 m blocks; route outcomes carry two stage bootstrap intervals (sites, then pairs). Site level claims use an exact sign flip test over the six sites, Holm corrected [37]; the smallest attainable two sided p is 1/32. On one core of a cloud x86 machine, the slip assessment of a 10 km^2 window takes 0.2 s at 30 m and 2.3 s at 10 m, and planning 100 routes on it 0.2 and 2.2 s. Every result in Tables III to V was registered with numeric predictions in a prediction log before it ran, but after the results of the first version on the same sites were known; the checks in Section IV-E are post hoc. The predictions of stages 25 to 28 (31 hits, 12 misses) and three amendments made before the affected results were known are in the supplement.

IV. RESULTS

A. Which layer sees faces

Table III gives the mean AUC of every layer for the face map label. On 10 m block means most layers rank face cells

TABLE II

HAZARD LAYERS. L: TRAINED ON THE LIDAR OF THE OTHER FIVE SITES (LEAVE ONE SITE OUT), ON ITS FACE LABELS OR, FOR SUPER RESOLUTION, ON ITS ELEVATION. EVERY LAYER THAT IS NOT A PROBABILITY IS CALIBRATED BY LOGISTIC REGRESSION [38] ON THE OTHER FIVE SITES WHEN A PLANNER NEEDS ONE.

Layer	Score of a cell	L
Slope	Horn slope of the coarse surface, the input of a slope costmap	
Ruggedness	Terrain ruggedness index [39] over the 8 neighbours	
Roughness	RMS residual of a plane fitted to 3×3 cells, as in GESTALT [14]	
Aspect rule	Slope times agreement of the cell's downslope azimuth with the dominant face direction of the site (from the coarse map)	
Interpolation	Steepest slope after cubic spline interpolation of the coarse surface to 1 m	
Super resolution	Steepest slope of a 5 m surface predicted by a dilated residual network from the interpolated 30 m surface	yes
Detector	Logistic model on slope, convexity, slope break and relief, trained on 60 000 synthetic straight crested dune profiles sampled at the cell size	
Boosting	Gradient boosted trees [40] on 14 terrain features (slope, ruggedness, roughness, curvature, topographic position, relief, detector features); variants with 5 more features (Boosting+), tuned by nested leave one site out, and without the detector features	yes
Network	Fully convolutional network [41], four 3×3 layers dilated by 1, 2, 4 and 8 [42], 1500 or 6000 steps	yes
+ imagery	Boosting on the terrain features plus Sentinel-2 or NAIP features (band means, brightness, contrast, shadow share); also imagery alone	yes

TABLE III

DETECTION OF THE FACE MAP LABEL (1): MEAN AUC OVER THE SIX SITES. I: LIDAR BLOCK MEANS; I10→30: THE LARGEST OF THE NINE 10 M SCORES IN EACH 30 M CELL; R10: USGS 1/3 ARC SECOND DEM; N30: NASADEM; G30: GLO-30; R10, N30 AND G30 ON THEIR OWN GRIDS. *TRAINED ON LIDAR LABELS OF THE OTHER FIVE SITES. BEST PER COLUMN IN BOLD (TIES AT TWO DECIMALS ALL BOLD); · NOT RUN.

Layer	I10	I30	I10→30	R10	N30	G30
Slope	0.90	0.69	·	0.90	0.62	0.65
Ruggedness	0.94	0.76	·	0.93	**0.63**	0.69
Roughness	0.80	0.77	·	·	·	·
Aspect rule	0.83	0.61	·	·	·	·
Interpolation	**0.97**	0.82	**0.96**	**0.96**	0.62	0.70
Super resolution*	·	0.84	·	·	·	·
Detector	0.95	0.74	0.95	0.95	0.56	0.65
Boosting*	**0.97**	0.81	**0.96**	0.95	**0.63**	**0.72**
Boosting+*	**0.97**	0.83	·	·	·	·
Boosting, tuned*	**0.97**	0.81	·	·	·	·
Network*	0.96	0.80	0.95	·	·	·
Network, 6000 steps*	·	0.84	·	·	·	·
Sentinel-2 only*	0.68	0.74	·	·	·	·
Boosting + Sentinel-2*	**0.97**	**0.85**	·	·	·	·
NAIP only*	0.62	0.67	·	·	·	·
Boosting + NAIP*	**0.97**	0.84	·	·	·	·

well (interpolation 0.97, boosting 0.97, the detector 0.95, slope 0.90), and the real USGS 1/3 arc second product behaves almost like the 10 m block mean (interpolation 0.96). At 30 m every elevation layer loses skill. With the default settings of the learned layers, interpolation is best (0.82), followed by boosting (0.81), the network (0.80), roughness (0.77) and ruggedness (0.76); the detector (0.74) is above only slope (0.69), NAIP alone (0.67) and the aspect rule (0.61). The learned layers were held back by those settings: learned super resolution of the 30 m surface reaches 0.84 and the network trained four times longer 0.84, the best elevation layers at 30 m, and boosting with five more features, among them the interpolated slope itself, 0.83. Tuning boosting by nested leave one site out changes nothing (0.81), nor does removing the detector's features (0.81). The information is still in the 10 m data: the largest of the nine 10 m interpolation scores in a 30 m cell reaches 0.96.

Imagery adds information that elevation lacks. Boosting on terrain plus Sentinel-2 features reaches 0.85 at 30 m (+0.04 over terrain alone), and with NAIP 0.84; imagery alone is weaker (0.74 and 0.67), and at 10 m it adds nothing (0.97). Super resolution and boosting with Sentinel-2 are above interpolation at all six sites and the detector below it at all six, but with six sites none of the seven comparisons against interpolation that we tested at the site level (detector, boosting, network, super resolution, boosting with curvature features, with Sentinel-2 and with NAIP) is significant after Holm correction (sign flip $p = 0.031$ at best, adjusted 0.22).

The real 30 m products are worse than the 30 m block mean: on their own grids the best layer reaches 0.72 on GLO-30 and 0.63 on NASADEM, which predates the lidar by about two decades; interpolating GLO-30 to the lidar grid changes little (best 0.72). Much of this loss is dune migration between the radar and lidar acquisitions rather than product error (Section IV-E).

The label matters for small faces. With the earlier cell label (at least 10 steep posts) the 30 m interpolation scores 0.82; requiring 100 or 300 steep posts raises it to 0.89 and 0.97, so a 30 m map ranks large faces well and small ones poorly; a 30 degree threshold changes little (0.82). The labels are spatially correlated over 30 to 870 m (the largest at Dumont, whose window holds rocky hills); with 870 m blocks the per site intervals of slope and the detector still lie below interpolation at all six sites.

B. How fine a map must be

Fig. 1a shows how detection falls with grid spacing. Every layer is almost perfect at 2 m (interpolation 1.00) and still good at 10 m (0.97); between 10 and 30 m interpolation falls to 0.82 and at 60 m to 0.69, while slope and the detector fall faster between 10 and 30 m. Fig. 1b relates the fall to face size, with the length of a patch taken as the major axis of an ellipse with the same second moments as its posts. At every spacing, boosting finds nearly all faces longer than four grid cells (96 % of faces 4 to 8 cells long at 30 m, 90 % at 60 m), but far fewer faces shorter than half a cell (60 % at 30 m, 46 % at 60 m). Longer patches touch more cells and so have more chances to be flagged, which accounts for part of this rise. Most faces in these fields are short at 30 m: 75 % of the patches are shorter than one 30 m cell.

C. Planning

Table IV gives mission success at the operating points chosen on the other five sites, and Fig. 2 shows the weight policy at every weight. With the posterior vehicle the gate routes at most 34 % of pairs at 10 m and 42 % at 30 m, whatever the layer, because the slip model is uncertain on most dune slopes (most refusals are at an endpoint: 76 % for boosting at 30 m and $\tau = 0.2$). The weight policy routes

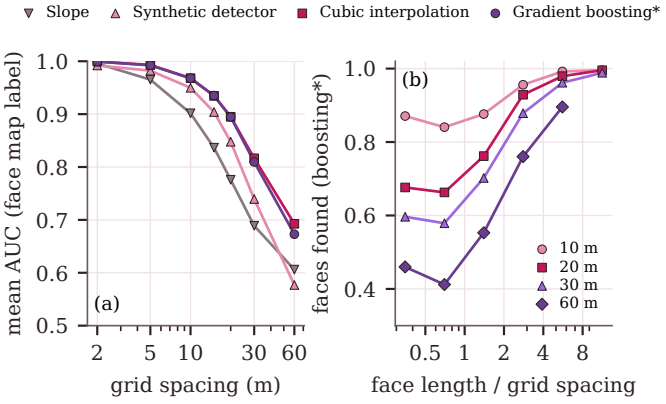


Fig. 1. Resolution and scale (lidar block means, face map label). (a) Mean AUC over six sites against the grid spacing. (b) Share of face patches found by boosting (a cell holding the patch is flagged at the threshold that flags 10 % of face free cells) against patch length over grid spacing, in bins with edges 0.5, 1, 2, 4 and 8, for four spacings; bins with fewer than 50 patches are omitted. Patch length: major axis of the ellipse with the same second moments. *Trained leave one site out.

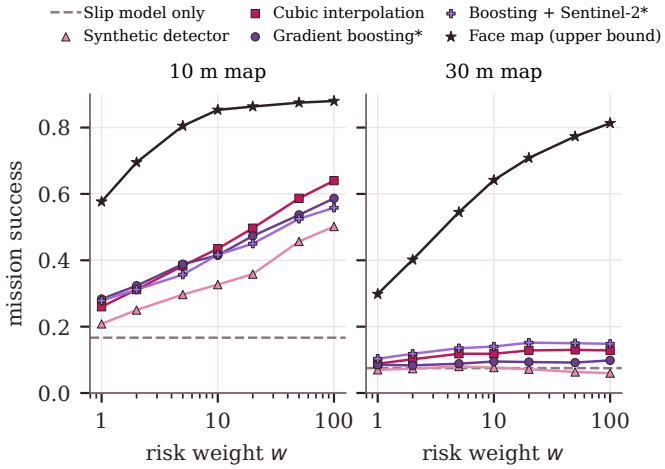


Fig. 2. Mission success of the weight policy against the risk weight w for each hazard layer (E1 endpoints, posterior vehicle, 600 pairs). Dashed: the slip model alone. The face map is an upper bound that needs face information from fine data. *Trained leave one site out.

almost every pair and is chosen for every layer; on 10 m maps it mostly takes the largest weight of the grid ($w = 100$).

On 10 m maps every layer helps. Mission success rises from 17 % with the slip model alone to 64 % with interpolation (+47 points, two stage interval +28 to +64, positive at all six sites), 59 % with boosting and 50 % with the detector. On 30 m maps the gain is small: interpolation adds +5 points (interval +2 to +9) and boosting with Sentinel-2 +5 (interval +1 to +9) to 8 %, and no layer computed from GLO-30 has an interval above zero. With the assumed vehicle the 30 m gain of interpolation is +7 points. Endpoints chosen without the truth (E2) lower every rate, the slip model alone to 3 % at 30 m, and there the gain of interpolation is no longer clearly above zero (interval 0 to +6).

The face map, an upper bound that assumes face information from fine data, reaches 81 % on the 30 m grid. The registered degraded face maps (Table IV, lower part, at $w = 5$; the gate routes too few pairs with this vehicle to compare) show how good that information must be: the face map gives 55 % at 30 m, displaced by 5 m 50 %, by 10 m 43 % and by 30 m 11 %. Face maps built with the same 28 degree rule from simulated 2 m and 5 m surveys give only 23 % and 8 %; Section IV-E shows why. The outcome is robust to the scoring details (supplement S5).

D. Onboard sensing

Table V adds onboard sensing that sees every face within the radius, an idealised bound. With $R = 20$ m the rover reaches the goal without touching a face on 80 % of the pairs when its 30 m prior holds only the slip model, and on 84 % with boosting as prior (+4 points, interval −1 to +13); the face map prior gives 85 %. No route between E1 endpoints touched a face and no pair used up its replan budget; every failure is a pair left without a route, either before the first move, when blocked cells and faces sensed at the start close every way

TABLE IV

MISSION SUCCESS (%) AT THE OPERATING POINT CHOSEN ON THE OTHER FIVE SITES (GATE THRESHOLD OR WEIGHT), 100 PAIRS PER SITE, POSTERIOR VEHICLE. E1: ENDPOINTS CHOSEN WITH THE TRUTH; E2: CHOSEN FROM THE COARSE MAP ONLY. †TWO STAGE BOOTSTRAP INTERVAL OF THE GAIN OVER THE SLIP MODEL ENTIRELY ABOVE ZERO. DEGRADED FACE MAPS: WEIGHT POLICY, $w = 5$. *TRAINED LEAVE ONE SITE OUT.

	I10		I30		G30
Layer	E1	E2	E1	E2	E1
Slip model only	17	9	8	3	6
Slope	35†	28†	6	3	6
Interpolation	64†	53†	13†	5	7
Detector	50†	33†	7	4	5
Detector, recalibrated	52†	37†	5	2	6
Boosting*	59†	47†	9	4	7
Network*	51†	40†	6	3	7
Boosting + Sentinel-2*	56†	46†	13†	7†	·
Face map (upper bound)	84†	80†	81†	68†	80†
Face map, $w = 5$	81	71	55	40	·
Face map 5 m off	67	56	50	39	·
Face map 10 m off	36	26	43	32	·
Face map 30 m off	19	10	11	5	·
Face map from 2 m survey	29	19	23	14	·
Face map from 5 m survey	17	9	8	4	·

inside the window (the same pairs for every prior: 4 of 120 pairs at 30 m, 17 at 10 m, which is why the 10 m prior does no better here), or after sensed faces closed the remaining ways. A better prior mainly saves replanning (29 replans per pair with the slip model prior, 8 with the face map). With E2 endpoints 5 % of the pairs touch a face whatever the prior, because some endpoints lie on or next to faces. A slip face lies behind a crest, out of sight of a rover climbing the windward slope (Section IV-E).

E. Post hoc checks

These checks were decided after a second review and are post hoc; pairs and the choice of operating point are as above.

TABLE V
CLOSED LOOP WITH ONBOARD SENSING (RADIUS R), 20 PAIRS PER SITE, POSTERIOR VEHICLE, SENSING EVERY FACE WITHIN THE RADIUS: SUCCESS (%; GOAL REACHED, NO FACE CONTACT) AND REPLANS PER PAIR (RP), E1 ENDPOINTS UNLESS MARKED E2. MAIN RUNS $R = 20$ M; I30 ALSO WITH $R = 10$ AND 30 M.

	I10, R 20		I30, R 20			I30, S	
Prior layer	S	Rp	S	Rp	E2	R 10	R 30
Slip model only	78	11	80	29	67	76	83
Detector	81	5	85	25	73	82	85
Interpolation	80	4	82	20	71	81	87
Boosting*	80	4	84	21	71	81	88
Face map	80	1	85	8	75	83	86

Line of sight. When the rover of Table V senses a 5 m cell only if the line from an eye 1 m above the ground to a point 0.2 m above the cell centre clears the lidar surface (I30, E1, $R = 20$ m, 120 pairs), success falls to 30 % with the slip model prior, and 65 % of the pairs touch a face. Boosting as prior gives 42 % (+12 points, interval +3 to +21, higher at all six sites) and the face map 74 % (+44 points, interval +25 to +63). Faces behind crests are seen too late, so the prior matters.

Finer and learned planning inputs. On 5 m block means, with weights up to $w = 1000$ (interpolation chose 1000 at every site), interpolation reaches 86 % against 34 % for the slip model alone (+52 points, interval +31 to +71) and the face map 93 %. On 30 m maps, the two layers that rank faces best add little: super resolution +5 points (interval +2 to +8) and the network trained for 6000 steps +2 (interval 0 to +5).

Displaced face maps at the chosen weight. With the weight chosen on the other five sites, the face map gives 84 % on the 30 m grid, displaced by 5 m 75 %, by 10 m 61 % and by 30 m 11 %.

Other checks (supplement S6). Scored only on faces of at least 300 posts, 30 m interpolation still adds only 13 points over the slip model, against 54 on 10 m maps; on a 5 m survey grid the 28 degree rule marks less than 60 % of the face posts at four sites; and dune migration, not only the posting, limits GLO-30 (AUC 0.85 at the two sites where it needs no shift, 0.62 elsewhere).

V. DISCUSSION AND LIMITATIONS

What the results mean. A 30 m elevation model keeps enough of the dune shape to rank face cells better than chance, but not enough to plan around faces: the best 30 m layers add a few points of mission success, while the same layers on a 10 m map add up to +47 points. Fig. 1b suggests why: faces several grid cells long are found at every spacing, and most faces in these fields are shorter than one 30 m cell. Better learning helps detection a little (super resolution 0.84 against 0.82 for interpolation) and planning about as much as interpolation (Section IV-E). Imagery gave the largest gain in 30 m detection, as in dune mapping [21], but not in planning. With a sensor that sees every face nearby the prior hardly matters, but faces hide behind crests, and with line of sight a good face map more than doubled success. The face map results put the remaining gap on information: face information from fine data, placed to within a few metres, would make most 30 m routes safe, so a drone survey must be georeferenced accordingly and its face threshold calibrated for its own resolution.

Arabian dunes. We did not test them. A face 100 m high at 32 degrees spans about 160 m downslope, five 30 m cells, so large megadune faces should be visible at 30 m and smaller flank faces not; but GLO-30 is weaker over sand [24] and dunes migrate.

Limitations. The truth is steep ground: rock, vehicle ruts and vegetated mounds also exceed 28 degrees, Dumont, Algodones and Little Sahara are off road vehicle areas, and the manual audit of a stratified sample of 600 patches is not finished, so label precision is unknown. We chose the windows, used them in earlier tests of the detector, and keep the prediction log in our own repository. The vehicle model is a posterior fitted to published slip curves of other vehicles; in our own leave one study out validation no slip model family transferred between published studies (median held out slip error 0.20 to 0.41 depending on the family), so absolute route outcomes are uncertain. No rover was driven, no physics was simulated, and even the line of sight check assumes that every visible face is detected. Six sites limit the training data of the learned layers and the power of site level tests.

VI. CONCLUSION

A 30 m elevation model ranks slip faces better than chance but cannot keep a small rover off them. On six lidar dune fields no hazard layer we planned with on such a map, including imagery fusion and learned super resolution, added more than a few points of mission success, while the same kind of layer on a 10 m map raised it from 17 % to 64 %, and the gap remained when only large faces were scored. Because faces hide behind crests, onboard sensing does not remove the need for a good prior: with line of sight a face map prior raised success from 30 % to 74 %. Next are the manual audit of the face labels, Arabian dune fields with a drone survey, and a rover whose slip has been measured on the same sand.

DATA AND CODE

USGS 3DEP lidar and DEMs, NASADEM and NAIP are in the public domain; Sentinel-2 data are provided under the Copernicus free licence; Copernicus GLO-30 is used under its licence (produced using Copernicus WorldDEM-30 © DLR e.V. 2010-2014 and © Airbus Defence and Space GmbH 2014-2018, provided under COPERNICUS by the European Union and ESA; all rights reserved). The code, result files, prediction log and the scripts that regenerate every number and figure of this paper and its supplement are at https://github.com/hudasol/dune-rover (stages 25 to 29).

REFERENCES

[1] European Space Agency and Airbus, "Copernicus DEM," Dataset, European Space Agency, 2022. [Online]. Available: https://doi.org/10.5270/ESA-c5d3d65

[2] K. Pye and H. Tsoar, *Aeolian Sand and Sand Dunes*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009.

[3] M. Heverly, J. Matthews, J. Lin, D. Fuller, M. Maimone, J. Biesiadecki, and J. Leichty, "Traverse performance characterization for the Mars Science Laboratory rover," *Journal of Field Robotics*, vol. 30, no. 6, pp. 835–846, 2013.

[4] R. E. Arvidson, K. D. Iagnemma, M. Maimone, A. A. Fraeman, F. Zhou, M. C. Heverly, P. Bellutta, D. Rubin, N. T. Stein, J. P. Grotzinger, and A. R. Vasavada, "Mars Science Laboratory Curiosity rover megaripple crossings up to sol 710 in Gale crater," *Journal of Field Robotics*, vol. 34, no. 3, pp. 495–518, 2017.

[5] W. Zhang and D. R. Montgomery, "Digital elevation model grid size, landscape representation, and hydrologic simulations," *Water Resources Research*, vol. 30, no. 4, pp. 1019–1028, 1994.

[6] N. Vandapel, R. R. Donamukkala, and M. Hebert, "Unmanned ground vehicle navigation using aerial ladar data," *The International Journal of Robotics Research*, vol. 25, no. 1, pp. 31–51, 2006.

[7] B. Sofman, E. Lin, J. A. Bagnell, J. Cole, N. Vandapel, and A. Stentz, "Improving robot navigation through self-supervised online learning," *Journal of Field Robotics*, vol. 23, no. 11-12, pp. 1059–1075, 2006.

[8] J. Delmerico, E. Mueggler, J. Nitsch, and D. Scaramuzza, "Active autonomous aerial exploration for ground robot path planning," *IEEE Robotics and Automation Letters*, vol. 2, no. 2, pp. 664–671, 2017.

[9] J.-M. Fortin, O. Gamache, W. Fecteau, E. Daum, W. Larrivée-Hardy, F. Pomerleau, and P. Giguère, "UAV-assisted self-supervised terrain awareness for off-road navigation," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 1–7.

[10] M. Ono, B. Rothrock, E. Almeida, A. Ansar, R. Otero, A. Huertas, and M. Heverly, "Data-driven surface traversability analysis for Mars 2020 landing site selection," in *Proc. IEEE Aerospace Conference*, 2016, pp. 1–12.

[11] G. Hedrick and Y. Gu, "Terrain-aware traverse planning for a Mars sample return rover," *Advanced Robotics*, vol. 35, no. 21–22, pp. 1286–1301, 2021.

[12] P. Papadakis, "Terrain traversability analysis methods for unmanned ground vehicles: A survey," *Engineering Applications of Artificial Intelligence*, vol. 26, no. 4, pp. 1373–1385, 2013.

[13] P. V. K. Borges, T. Peynot, S. Liang, B. Arain, M. Wildie, M. G. Minareci, S. Lichman, G. Samvedi, I. Sa, N. Hudson, M. Milford, P. Moghadam, and P. Corke, "A survey on terrain traversability analysis for autonomous ground vehicles: Methods, sensors, and challenges," *Field Robotics*, vol. 2, pp. 1567–1627, 2022.

[14] S. B. Goldberg, M. W. Maimone, and L. Matthies, "Stereo vision and rover navigation software for planetary exploration," in *Proceedings, IEEE Aerospace Conference*, vol. 5, Big Sky, MT, USA, 2002, pp. 5–2025–5–2036.

[15] M. Benrabah, C. O. Mousse, E. Randriamiarintsoa, R. Chapuis, and R. Aufrère, "A review on traversability risk assessments for autonomous ground vehicles: Methods and metrics," *Sensors*, vol. 24, no. 6, p. 1909, 2024.

[16] G. Ishigami, A. Miwa, K. Nagatani, and K. Yoshida, "Terramechanics-based model for steering maneuver of planetary exploration rovers on loose soil," *Journal of Field Robotics*, vol. 24, no. 3, pp. 233–250, 2007.

[17] S. Shrivastava, A. Karsai, Y. O. Aydin, R. Pettinger, W. Bluethmann, R. O. Ambrose, and D. I. Goldman, "Material remodeling and unconventional gaits facilitate locomotion of a robophysical rover over granular terrain," *Science Robotics*, vol. 5, no. 42, p. eaba3499, 2020.

[18] H. Inotsume, T. Kubota, and D. Wettergreen, "Robust path planning for slope traversing under uncertainty in slip prediction," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3390–3397, 2020.

[19] M. Endo, T. Taniai, R. Yonetani, and G. Ishigami, "Risk-aware path planning via probabilistic fusion of traversability prediction for planetary rovers on heterogeneous terrains," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 11 852–11 858.

[20] M. Ono, T. J. Fuchs, A. Steffy, M. Maimone, and J. Yen, "Risk-aware planetary rover operation: Autonomous terrain classification and path planning," in *2015 IEEE Aerospace Conference*, Big Sky, MT, 2015, pp. 1–10.

[21] C. H. Hugenholtz, N. Levin, T. E. Barchyn, and M. C. Baddock, "Remote sensing and spatial analysis of aeolian sand dunes: A review and outlook," *Earth-Science Reviews*, vol. 111, no. 3–4, pp. 319–334, 2012.

[22] J. Daynac, P. Bessin, S. Pochat, R. Mourgues, and S. Shumack, "A new workflow for mapping dune features (outline, crestline and defects) combining deep learning and skeletonization from DEM-derived data," *Geomorphology*, vol. 463, p. 109369, 2024.

[23] P. L. Guth and T. M. Geoffroy, "LiDAR point cloud and ICESat-2 evaluation of 1 second global digital elevation models: Copernicus wins," *Transactions in GIS*, vol. 25, no. 5, pp. 2245–2261, 2021.

[24] M. Martone, B. Bräutigam, P. Rizzoli, N. Yague-Martinez, and G. Krieger, "Enhancing interferometric SAR performance over sandy areas: Experience from the TanDEM-X mission," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 9, no. 3, pp. 1036–1046, 2016.

[25] M. Wang, H. Li, Y. Liu, and H. Li, "Multi-source DEM vertical accuracy evaluation of Taklimakan desert hinterland based on ICESat-2 ATL08 and UAV data," *Remote Sensing*, vol. 17, no. 11, p. 1807, 2025.

[26] Z. Chen, X. Wang, Z. Xu, and W. Hou, "Convolutional neural network based DEM super resolution," in *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLI-B3, 2016, pp. 247–250.

[27] X. Han, X. Ma, H. Li, and Z. Chen, "A global-information-constrained deep learning network for digital elevation model super-resolution," *Remote Sensing*, vol. 15, no. 2, p. 305, 2023.

[28] L. J. Sugarbaker, E. W. Constance, H. K. Heidemann, A. L. Jason, V. Lukas, D. L. Saghy, and J. M. Stoker, "The 3D Elevation Program initiative: A call for action," U.S. Geological Survey, Circular 1399, 2014.

[29] B. K. P. Horn, "Hill shading and the reflectance map," *Proceedings of the IEEE*, vol. 69, no. 1, pp. 14–47, 1981.

[30] U.S. Geological Survey, "1/3 arc second digital elevation models, 3D Elevation Program," Dataset, The National Map, 2026, accessed October 2026. [Online]. Available: https://apps.nationalmap.gov/downloader/

[31] NASA JPL, "NASADEM merged DEM global 1 arc second V001," Dataset, NASA EOSDIS Land Processes Distributed Active Archive Center, 2020. [Online]. Available: https://doi.org/10.5067/MEaSUREs/NASADEM/NASADEM_HGT.001

[32] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sensing of Environment*, vol. 120, pp. 25–36, 2012.

[33] U.S. Department of Agriculture, Farm Service Agency, "National agriculture imagery program (NAIP)," Dataset, through the Microsoft Planetary Computer, 2026, accessed October 2026. [Online]. Available: https://naip-usdaonline.hub.arcgis.com/

[34] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.

[35] J. Y. Wong, *Theory of Ground Vehicles*, 4th ed. Wiley, 2008.

[36] H. R. Künsch, "The jackknife and the bootstrap for general stationary observations," *The Annals of Statistics*, vol. 17, no. 3, pp. 1217–1241, 1989.

[37] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979. [Online]. Available: https://www.jstor.org/stable/4615733

[38] J. C. Platt, "Probabilities for SV machines," in *Advances in Large-Margin Classifiers*, A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans, Eds. Cambridge, MA: The MIT Press, 2000, pp. 61–74.

[39] S. J. Riley, S. D. DeGloria, and R. Elliot, "A terrain ruggedness index that quantifies topographic heterogeneity," *Intermountain Journal of Sciences*, vol. 5, no. 1-4, pp. 23–27, 1999. [Online]. Available: https://arc.lib.montana.edu/ojs/index.php/IJS/article/view/2914

[40] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, no. 85, pp. 2825–2830, 2011. [Online]. Available: https://jmlr.org/papers/v12/pedregosa11a.html

[41] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 3431–3440.

[42] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *International Conference on Learning Representations (ICLR)*, 2016. [Online]. Available: https://arxiv.org/abs/1511.07122